

Dust Calibration System

Abstract

The utility model provides a dust calibration system, including monodispersity particle generator, monodispersity particle generator passes through the pipeline and is connected with the one end of static neutralizer, the other end of static neutralizer passes through the pipeline and is connected with the homogeneous mixing case, the even very lower extreme of mould assembling is connected with the shunt, the shunt respectively with the entry linkage of PM10 cutting head and TSP cutting head. Dust that this calibration system is fit for various granules is like dust on the roads, ISO dirt, SAE dirt, PSL dry powder etc. And the measuring result deviation is little.

Description

translated from Chinese

Dust Calibration System

FIELD

[0001] The present invention relates to a dust calibration system.

Background technique

[0002] Currently, the digital dust instrument has been widely used in indoor air quality testing, workplace air quality testing, mine dust concentration detector and outdoor air quality testing. Its different manufacturers produce different naming dust instrument, such as a digital dust meter, intelligent digital dust meter, micro dust instrument, meter respirable dust, dust explosion-like instrument. In general, these instruments can be collectively referred to as dust instrument, detecting the concentration of dust particles in the ambient air quality of the instrument. Dust Monitor The measuring principle can be divided light scattering instrument and the piezoelectric Dust Dust Monitor two kinds balance formula. Instrument light scattering dust scattering laser light flux of dust concentration was measured according to the dust particles, such instruments relatively simple construction, fast response, easy maintenance, digital dust meter for the current mainstream products, more than 90% of the total market . However, light scattering dust analyzer manufacturers used a light source and a light detector chamber vary, before using the instrument factory calibration methods vary, resulting in the instrument response curve and accuracy vastly different Wan another IJ, measured under the same environment quality difference larger concentration of inconvenience to users, data comparability poor.

SUMMARY

[0003] For the above disadvantages and problems of the prior art, the present invention aims to provide a good comparability dust calibration system.

[0004] To achieve the above object, the present invention provides the following technical solutions:

[0005] A dust calibration system, comprising a generator of monodisperse particles, the monodisperse particle generator connected to one end of the conduit with an electrostatic filter and the other end of the electrostatic filter and uniformly mixing tank by piping connection the uniform mixing tank is connected to the lower end of the shunt, the shunt are connected to the cutting head and the inlet PMIO TSP cutting head;

[0006] The PMIO cutting head mounted at the outlet of the first clip holder, the clip is mounted on the first bracket has a first filter, the first clamp holder with a standard meter is connected to the flow meter and the second standard Li a connector;

[0007] The TSP is installed at the outlet of the cutting head second clip holder, a second filter mounted on said second clamp holder, the second clamp holder is connected to a standard flow meter, the flow meter and the second standard two chestnut connection.

[0008] Further, interposed between the first holder and the second clamp meter with the standard tray with a standard flowmeter has a control valve.

[0009] Further, the uniform mixing tank and monodisperse ion generator has a high efficiency air filter.

[0010] Further, the first membrane and second membrane is a membrane 7mmPTFE.

[0011] The beneficial effects of the present invention, the present invention since the monodisperse particle generator is installed, so that the system is suitable for a variety of road dust particles such as dust, ISO Dust, SAE dust, the PSL dry powder, the particle size range : 0.1~150 μm, the particle concentration inside: 0.02~100mg / m3 continuously adjustable, stability: two measurements the deviation <5%.

BRIEF DESCRIPTION

[0012] In order to more clearly illustrate the embodiments of the present invention or the technical solution in the prior art, accompanying drawings for describing the embodiments or the prior art described in the introduction required simply Apparently, the description below the drawings are only some embodiments of the present invention embodiments, those of ordinary skill in the art is concerned, without any creative effort, and can obtain other drawings based on these drawings.

[0013] FIG. 1 is a schematic view of the present invention the dust calibration system.

Detailed ways

[0014] Next, the present invention in conjunction with the accompanying drawings, of technical solution of the present invention will be clearly and completely described, obviously, the described embodiments are merely part of embodiments of the present invention rather than all embodiments. Based on the present invention in embodiments, those of ordinary skill in the art to make all other embodiments without creative work obtained, are within the scope of protection of the present invention.

[0015] As shown, the present invention discloses a dust calibration system 1, the generator comprising monodisperse particles, the monodisperse particle generator connected to one end of the conduit and the electrostatic filter, the electrostatic and the other end of the conduit is uniform through the mixing box connected to a lower end of a very uniform bonding box is connected to the shunt, the shunt are connected to the cutting head and the inlet PMIO TSP cutting head;

[0016] The PMIO cutting head mounted at the outlet of the first holder clip, the first clip is attached to the first clip first filter holder connected to the holder with a standard meter, the flow meter with a first standard Li is connected;

[0017] The TSP is installed at the outlet of the cutting head second clamp holder is attached to a second membrane and the second clamp holder with a standard meter is connected to said second clamp holder, and the second standard flowmeters chestnut connection.

[0018] The preferred technical solution, the first control valve interposed between each tray is provided with a second flow meter and a standard clamp meter with the standard tray.

[0019] The preferred technical solution, the mixing tank and uniformly dispersing the single ion generator equipped with HEPA filters.

[0020] The preferred technical solution, the first membrane and second membrane is a membrane 7mmPTFE.

[0021] The calibration system requires at least five kinds of single dispersion particle diameter of the PSL particles (e.g. 5 microns and 8 microns, 10 microns, 12 microns, 15 microns, etc.). Samples of each powder of particle diameter, it needs to take place with a low dose of dust diffuser, through electrostatic neutralizer after uniformly mixed into the tank. The aerosol particles mixed with clean dilution air are uniformly mixed in the tank, out of the collected isokinetic sampling head. Aerosol particles taken out from the test tube into the two-way shunt inlet and PM-10 inlet TSP respectively, two parts by the flow control valve is set at a nominal flow rate meter dust, aerosol particles are collected in two 47mm PTFE on a filter, the same acquisition time at the same flow rate, and then remove the filter for weighing. Cutting efficiency ratio by weight of the particles collected by the filter is very two PM-10 inlet. Cutting efficiency were made PM-10 particles sizes inlet 5, the cutting curve obtained, corresponding to 50% of the particle diameter is the cutting efficiency of the cutting point, it determines whether the cut points with the nominal cutting instrument consistent point.

[0022] The above are only the present invention specific embodiments, the scope of protection of the present invention is not limited thereto, any skilled in the art in the art within the technical scope disclosed in the present invention can be easily variations or replacement that, shall fall within the scope of the present invention. Accordingly, the scope of protection of the present invention should be the scope of the claims.

Cited By (1)

Publication number	Priority date	Publication date	Assignee	Title
CN104111217A *	2014-06-18	2014-10-22	瑞曼博(北京)环保科技有限公司	Dust Calibration System

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

Publication	Publication Date	Title
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Li et al.	2000	Evaluation of six inhalable aerosol samplers
US7029921B2	2006-04-18	Apparatus for rapid measurement of aerosol bulk chemical composition
Görner et al.	2001	Study of fifteen respirable aerosol samplers used in occupational hygiene
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Lundgren et al.	1975	The mass distribution of large atmospheric particles
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Tan et al.	2013	Design and application of an unattended multifunctional H-TDMA system
LIPPMANN et al.	1974	Calibration of dual-inlet cyclones for "respirable" mass sampling

Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title
CN201420323998.4U	2014-06-18	2014-06-18	Dust Calibration System

Applications Claiming Priority (1)

Application	Filing date	Title
CN201420323998.4U	2014-06-18	Dust Calibration System

Legal Events

Date	Code	Title	Description
2015-11-25	C14	Grant of patent or utility model	
2019-03-12	DD01	Delivery of document by public notice	
2019-08-09	CF01	Termination of patent right due to non-payment of annual fee	

Concepts

machine-extracted

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Name	Image	Sections	Count	Query match
dust		abstract	3	0
cutting process		abstract	2	0
particles		abstract	2	0
static		abstract	2	0
granular material		abstract	1	0
mixing		abstract	1	0
powders		abstract	1	0
trisodium phosphate		abstract	1	0

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Other languages: Chinese

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Worldwide applications

2014 - CN

Application CN201420323998.4U events

Ⓢ

2014-06-18

Application filed by 瑞曼博(北京)环保科技有限公司

2014-06-18

Priority to CN201420323998.4U

2015-11-25

Application granted

2015-11-25

Publication of CN204807424U

Info: Cited by (1), Legal events, Similar documents, Priority and Related Applications

External links: Espacenet, Global Dossier, Discuss

Claims (4)

Hide Dependent

translated from Chinese

A dust calibration system, characterized in that the generator comprising monodisperse particles, the monodisperse particle generator connected to one end of the conduit with an electrostatic filter and the other end of the electrostatic filter and through a pipe with a uniform connected to the mixing tank, the tank is uniformly mixed with a lower end connected to the splitter, the splitter respectively PMIO TSP cutting head and the cutting head inlet connection; PMIO the cutting head is attached to the first outlet of the clip holder, the first a clamp bracket mounted on the first membrane, said first clamping holder is connected to a standard flow meter, the flow meter with a first standard Li; said TSP cutting head mounted at the outlet of the second clamp holder, said mounting the second clamp holder with a second membrane, said second clamp holder is connected to a standard flow meter, a flow meter and a second standard connector Li.

2. dust calibration system according to claim 1, wherein said first control valve interposed between each tray is provided with a standard flowmeter and the flowmeter standard and a second clamp holder.

3. Dust calibration system according to claim 1, wherein said tank and uniformly mixing the monodisperse ion generator has high efficiency air filter.

4. The dust calibration system of claim 1, wherein said first membrane and second membrane is a membrane 7mmPTFE.